LESSON PLAN

PISTON ENGINE (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	IC Engines, Variations and usage in aviation industry as main propulsion plant and auxiliary machinery, operation cycles of four stroke and
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	IC Engines, Variations and usage in aviation industry as main propulsion plant and auxiliary machinery, operation cycles of four stroke and two stroke Piston engine, comparison, advantages, modified Otto cycle
5. Specific Objectives	15.1.1.1 Trainees will be able to explain about the working principle of piston engines.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of Piston Engine Diagrams, Safety: Handle diagrams carefully, Instructor supervision: Monitor understanding of cycle phases
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an internal combustion engine?
(b) Name two types of propulsion plants used in aviation.
(c) Define the term 'auxiliary machinery' in an aircraft context.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Working principle of piston engines • Four stroke operation cycle phases • Two stroke operation cycle phases • Comparison of four and two stroke engines • Advantages of two stroke engines	L & Ds	PPT, IB	15 min
(b)	• Usage as main propulsion plant • Usage as auxiliary machinery • Modified Otto cycle explanation • Variations in aviation piston engines • Efficiency comparison of cycles	L & Ds	PPT, PDF	20 min
(c)	• Summary of engine cycles • Key differences recap • Application scenarios • Review of specific objectives • Preparation for next lesson	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between the intake stroke in four and two stroke engines.
(b) Describe one advantage of using a modified Otto cycle in aviation applications.

12. Home Assignment and Follow Up	Prepare a brief note comparing the fuel consumption characteristics of four stroke and two stroke engines for use in LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	IC Engines, Variations and usage in aviation industry as main propulsion plant and auxiliary machinery, operation cycles of four stroke and two stroke Piston engine, comparison, advantages, modified Otto cycle	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min